

AGREEGATE FUNCTIONS:

Program:

```
1  -- Step 1: Create Database and Table
2  ● CREATE DATABASE aggregate_bank_db;
3  ● USE aggregate_bank_db;
4  ● CREATE TABLE bank_transactions (
5      txn_id INT PRIMARY KEY,
6      customer_name VARCHAR(50),
7      branch_name VARCHAR(50),
8      transaction_type VARCHAR(20),
9      amount DECIMAL(10,2),
10     transaction_date DATE
11 );
12  -- Step 2: Insert Data
13  ● INSERT INTO bank_transactions VALUES
14     (101,'Tejaswini','Hyderabad','Deposit',5000,'2024-01-05'),
15     (102,'Chinni','Hyderabad','Withdrawal',2000,'2024-01-06'),
16     (103,'Sree','Vijayawada','Deposit',12000,'2024-01-08'),
17     (104,'Latha','Vizag','Deposit',8000,'2024-01-10'),
18     (105,'Ravi','Hyderabad','Withdrawal',3500,'2024-01-11'),
19     (106,'Praneeth','Vizag','Deposit',15000,'2024-01-12'),
20     (107,'Sahasra','Vijayawada','Withdrawal',1000,'2024-01-13'),
21     (108,'Bhavani','Hyderabad','Deposit',9000,'2024-01-14'),
22     (109,'Nikila','Vizag','Withdrawal',4000,'2024-01-15'),
23     (110,'Karthik','Vijayawada','Deposit',11000,'2024-01-16');
```

Output:

Output					
Action Output					
#	Time	Action	Message	Duration / Fetch	
✓ 1	15:03:45	CREATE DATABASE aggregate_bank_db	1 row(s) affected	0.015 sec	
✓ 2	15:03:47	USE aggregate_bank_db	0 row(s) affected	0.000 sec	
✓ 3	15:03:51	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, ...	0 row(s) affected	0.031 sec	
✓ 4	15:04:02	INSERT INTO bank_transactions VALUES (101,'Tejaswini','Hyderaba...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.015 sec	

Program:

```
24      -- Step 3: SUM()
25      SELECT SUM(amount) AS Total_Amount
26      FROM bank_transactions;
```

Output:

Result Grid	
	Total_Amount
▶	70500.00

5 15:06:07 SELECT SUM(amount) AS Total_Amount FROM bank_transactions ... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```
27      -- Step 4: AVG()
28      SELECT AVG(amount) AS Average_Transaction
29      FROM bank_transactions;
```

Output:

Result Grid	
	Average_Transaction
▶	7050.000000

6 15:06:49 SELECT AVG(amount) AS Average_Transaction FROM bank_transa... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```
30      -- Step 5: MAX()
31      SELECT MAX(amount) AS Highest_Transaction
32      FROM bank_transactions;
```

Output:

Result Grid	
	Highest_Transaction
▶	15000.00

7 15:07:33 SELECT MAX(amount) AS Highest_Transaction FROM bank_trans... 1 row(s) returned 0.016 sec / 0.000 sec

Program:

```
33      -- Step 6: MIN()
34      SELECT MIN(amount) AS Lowest_Transaction
35      FROM bank_transactions;
```

Output:

Result Grid		Filter Rows:
	Lowest_Transaction	
▶	1000.00	

✓	8 15:08:20	SELECT MIN(amount) AS Lowest_Transaction FROM bank_trans...	1 row(s) returned	0.000 sec / 0.000 sec
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Program:

```

36      -- Step 7: COUNT()
37 •    SELECT COUNT(*) AS Total_Transactions
38      FROM bank_transactions;
```

Output:

Result Grid		Filter Rows:
	Total_Transactions	
▶	10	

✓	9 15:09:12	SELECT COUNT(*) AS Total_Transactions FROM bank_transactio...	1 row(s) returned	0.000 sec / 0.000 sec
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Program:

```

39      -- Step 8: WHERE + SUM()
40 •    SELECT SUM(amount) AS Total_Deposit
41      FROM bank_transactions
42      WHERE transaction_type = 'Deposit';
```

Output:

Result Grid		Filter Rows:
	Total_Deposit	
▶	60000.00	

✓	10 15:09:55	SELECT SUM(amount) AS Total_Deposit FROM bank_transaction...	1 row(s) returned	0.000 sec / 0.000 sec
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Program:

```

43      -- Step 9: GROUP BY + SUM()
44 •    SELECT branch_name,
45           SUM(amount) AS Total_Amount
46      FROM bank_transactions
47      GROUP BY branch_name;
```

Output:

Result Grid			Filter Rows:	
	branch_name	Total_Amount		
▶	Hyderabad	19500.00		
	Vijayawada	24000.00		
	Vizag	27000.00		

✓	11	15:10:36	SELECT branch_name, SUM(amount) AS Total_Amount FRO...	3 row(s) returned	0.000 sec / 0.000 sec
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Program:

```

48      -- Step 10: GROUP BY + HAVING
49  ●    SELECT branch_name,
50          SUM(amount) AS Total_Amount
51      FROM bank_transactions
52      GROUP BY branch_name
53      HAVING SUM(amount) > 20000;

```

Output:

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Result Grid

Form Editor

Field Types

Query Stats

	branch_name	Total_Amount
▶	Vijayawada	24000.00
	Vizag	27000.00

Result 8

Read Only

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 7	15:07:33	SELECT MAX(amount) AS Highest_Transaction FROM bank_trans...	1 row(s) returned	0.016 sec / 0.000 sec
✓ 8	15:08:20	SELECT MIN(amount) AS Lowest_Transaction FROM bank_trans...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 9	15:09:12	SELECT COUNT(*) AS Total_Transactions FROM bank_transactio...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 10	15:09:55	SELECT SUM(amount) AS Total_Deposit FROM bank_transaction...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 11	15:10:36	SELECT branch_name, SUM(amount) AS Total_Amount FRO...	3 row(s) returned	0.000 sec / 0.000 sec
✓ 12	15:12:12	SELECT branch_name, SUM(amount) AS Total_Amount FRO...	2 row(s) returned	0.000 sec / 0.000 sec

Program:

```
54      -- Step 11: GROUP BY + ORDER BY
55      ● SELECT branch_name,
56             SUM(amount) AS Total_Amount
57      FROM bank_transactions
58      GROUP BY branch_name
59      ORDER BY Total_Amount DESC;
```

Output:

	branch_name	Total_Amount
▶	Vizag	27000.00
	Vijayawada	24000.00
	Hyderabad	19500.00

✓ 13 15:12:51 SELECT branch_name, SUM(amount) AS Total_Amount FRO... 3 row(s) returned 0.000 sec / 0.000 sec

Program:

```
60      -- Step 12: GROUP BY + HAVING + ORDER BY
61      ● SELECT branch_name,
62             COUNT(*) AS Total_Transactions
63      FROM bank_transactions
64      GROUP BY branch_name
65      HAVING COUNT(*) >= 3
66      ORDER BY Total_Transactions DESC;
```

Output:

	branch_name	Total_Transactions
▶	Hyderabad	4
	Vijayawada	3
	Vizag	3

✓ 14 15:14:24 SELECT branch_name, COUNT(*) AS Total_Transactions FR... 3 row(s) returned 0.000 sec / 0.000 sec

Program:

```
67      -- Step 13: WHERE + COUNT()
68  ●    SELECT COUNT(*) AS Withdrawals
69      FROM bank_transactions
70      WHERE transaction_type = 'Withdrawal';
```

Output:

Result Grid	
Withdrawals	
▶	4

15 15:15:19 SELECT COUNT(*) AS Withdrawals FROM bank_transactions WH... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```
71      -- Step 14: WHERE + AVG()
72  ●    SELECT AVG(amount) AS Avg_Deposit
73      FROM bank_transactions
74      WHERE transaction_type = 'Deposit';
```

Output:

Result Grid	
Avg_Deposit	
▶	10000.000000

16 15:16:06 SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions ... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```
75      -- Step 15: GROUP BY + MAX()
76  ●    SELECT branch_name,
77          MAX(amount) AS Highest_Amount
78      FROM bank_transactions
79      GROUP BY branch_name;
```

Output:

Result Grid		Filter Rows:
branch_name Highest_Amount		
▶	Hyderabad	9000.00
	Vijayawada	12000.00
	Vizag	15000.00

✓ 17 15:17:51 SELECT branch_name, MAX(amount) AS Highest_Amount FR... 3 row(s) returned 0.000 sec / 0.000 sec

Program:

```
80 -- Step 16: GROUP BY + MIN()
81 ● SELECT branch_name,
82      MIN(amount) AS Lowest_Amount
83 FROM bank_transactions
84 GROUP BY branch_name;
```

Output:

	branch_name	Lowest_Amount
▶	Hyderabad	2000.00
	Vijayawada	1000.00
	Vizag	4000.00

✓ 18 15:18:48 SELECT branch_name, MIN(amount) AS Lowest_Amount FR... 3 row(s) returned 0.000 sec / 0.000 sec

Program:

```
85 -- Step 17: WHERE + ORDER BY
86 ● SELECT *
87 FROM bank_transactions
88 WHERE amount > 3000
89 ORDER BY amount DESC;
```

Output:

Result Grid						
bxn_id	customer_name	branch_name	transaction_type	amount	transaction_date	
106	Praneeth	Vizag	Deposit	15000.00	2024-01-12	
103	Sree	Vijayawada	Deposit	12000.00	2024-01-08	
110	Karthik	Vijayawada	Deposit	11000.00	2024-01-16	
108	Bhavani	Hyderabad	Deposit	9000.00	2024-01-14	
* HIDE	HIDE	HIDE	HIDE	HIDE	HIDE	

#	Time	Action	Message	Duration / Fetch
✓ 14	15:14:24	SELECT branch_name, COUNT(*) AS Total_Transactions FR...	3 row(s) returned	0.000 sec / 0.000 sec
✓ 15	15:15:19	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WH...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 16	15:16:06	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions ...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 17	15:17:51	SELECT branch_name, MAX(amount) AS Highest_Amount FR...	3 row(s) returned	0.000 sec / 0.000 sec
✓ 18	15:18:48	SELECT branch_name, MIN(amount) AS Lowest_Amount FR...	3 row(s) returned	0.000 sec / 0.000 sec
✓ 19	15:19:39	SELECT * FROM bank_transactions WHERE amount > 8000 OR...	4 row(s) returned	0.000 sec / 0.000 sec

Program:

```

90      -- Step 18: GROUP BY + COUNT()
91      SELECT branch_name,
92             COUNT(*) AS Customer_Count
93      FROM bank_transactions
94      GROUP BY branch_name;

```

Output:

Result Grid		
branch_name	Customer_Count	
Hyderabad	4	
Vijayawada	3	
Vizag	3	

✓ 20	15:20:28	SELECT branch_name, COUNT(*) AS Customer_Count FROM...	3 row(s) returned	0.000 sec / 0.000 sec
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Program:


```

95      -- Step 19: HAVING + COUNT()
96  ●   SELECT branch_name,
97         COUNT(*) AS Total_Transactions
98      FROM bank_transactions
99      GROUP BY branch_name
100     HAVING COUNT(*) > 3;

```

Output:

Result Grid			Filter Rows:
	branch_name	Total_Transactions	
▶	Hyderabad	4	

21 15:21:16 SELECT branch_name, COUNT(*) AS Total_Transactions FR... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```

101     -- Step 20: GROUP BY + AVG()
102  ●   SELECT branch_name,
103         AVG(amount) AS Avg_Amount
104     FROM bank_transactions
105     GROUP BY branch_name;

```

Output:

Result Grid			Filter Rows:
	branch_name	Avg_Amount	
▶	Hyderabad	4875.000000	
	Vijayawada	8000.000000	
	Vizag	9000.000000	

22 15:21:59 SELECT branch_name, AVG(amount) AS Avg_Amount FROM... 3 row(s) returned 0.015 sec / 0.000 sec

Program:

```

106     -- Step 21: WHERE + GROUP BY + HAVING
107  ●   SELECT branch_name,
108         SUM(amount) AS Total_Deposit
109     FROM bank_transactions
110     WHERE transaction_type = 'Deposit'
111     GROUP BY branch_name
112     HAVING SUM(amount) > 15000;

```

Output:

Result Grid  Filter Rows:		
	branch_name	Total_Deposit
▶	Vijayawada	23000.00
	Vizag	23000.00

✓ 23 15:22:47 SELECT branch_name, SUM(amount) AS Total_Deposit FRO... 2 row(s) returned 0.000 sec / 0.000 sec

Program:

```
113      -- Step 22: GROUP BY + HAVING + ORDER BY
114  ●    SELECT branch_name,
115         AVG(amount) AS Avg_Amount
116      FROM bank_transactions
117      GROUP BY branch_name
118      HAVING AVG(amount) > 7000
119      ORDER BY Avg_Amount DESC;
```

Output:

Result Grid  Filter Rows:		
	branch_name	Avg_Amount
▶	Vizag	9000.000000
	Vijayawada	8000.000000

✓ 24 15:23:29 SELECT branch_name, AVG(amount) AS Avg_Amount FROM... 2 row(s) returned 0.000 sec / 0.000 sec